

Individual Work Report

Benchmark for Structured Information Processing in LLMs - FSS 2026

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Sprint / Period	Sprint 4
Team Leader this Sprint	Nishit Suthar
Submission Date	4 th August

1. My Contributions This Sprint

SL.No.	Contribution / Task
1	identified 185 candidate Q&A pairs (20 Sprint 2 structured failures + 165 Sprint 3 UDA failures) and established the criterion that only questions failing consistently across models are included.
2	Executed evaluation across all 185 questions using fixed config (TEMP=0.0, CHUNK=3000, OVERLAP=300, TOP_K=5, all-MiniLM-L6-v2 embeddings), 184/185 answered, yielding 76 failures (31 finhybrid, 29 tathybrid, 16 music_structured).
3	Categorized all 76 failures using rule-based logic into 10 failure categories — top two: Financial (29 questions, 38%), Set operations / Decade comparison (24 questions, 32%).
4	Extended the RAG framework to support OpenRouter as a new provider; added two new models e.g. Nemotron-550B (OpenRouter) and Mistral Small-2603, with reasoning enabled via the "reasoning": {"enabled": True} flag.
5	Ran both new models on the 76 Gemini failure questions; documented cross-model comparison: 75/76 questions (98.7%) failed all three models, confirming they constitute a robust hard-cases benchmark.
6	Built a master comparison table (benchmark_comparison_all_models.csv, 76 rows × 3 models) and aggregated scores summary (benchmark_scores_summary.csv) covering answer rate, accuracy, and per-dataset breakdown for all three models.
7	Developed a self-contained interactive HTML benchmark dashboard (benchmark_dashboard.html, ~81 KB) featuring 8 KPI cards, 6 Chart.js charts, filterable question table, per-row modal with full question/answers, and PDF export (Server Less).

2. Joint Contributions

Task / Contribution	All Contributors	My Role in Joint Work

No Joint Work